

Positive Square Energy of Every Hexacyclic Cactus

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Abstract

Let G be a connected hexacyclic cactus of order n . We prove $s^+(G) > n$. Exact block-additive doubly nonnegative optimization leaves only the cycle multisets $\{3, 3, 3, 3, 3, q\}$ and $\{3, 3, 3, 3, 5, 5\}$. For disconnected shared-cut graphs we prove a leaf-cluster theorem for the first family and audit all twenty-eight colored cluster partitions for the second. The two entry-sensitive rows are closed by a six-row component classification and by an exact census of eight incidences and twenty-six ordered entry orbits. For one shared-cut cluster, exact color-preserving incidence censuses contain respectively 68, 70, 71 trees in the three q -capacity regimes and 150 trees in the four-triangle–two-pentagon family. All ordinary splits are certified; the bouquets and the unique saturated pentagon hub are handled directly. Every construction permits arbitrary bridge connectors and arbitrary trees attached at arbitrary vertices.

1 Preliminaries and dependencies

All graphs are finite, simple, and undirected. A cactus is a connected graph whose blocks are bridges or cycles. Its cyclomatic number is the number of cyclic blocks. If the adjacency eigenvalues of G are $\lambda_1, \dots, \lambda_n$, put

$$s^+(G) = \sum_{\lambda_j > 0} \lambda_j^2, \quad s^-(G) = \sum_{\lambda_j < 0} \lambda_j^2, \quad \sigma(G) = s^+(G) - |V(G)|.$$

For every vertex partition $V(G) = V_1 \dot{\cup} \dots \dot{\cup} V_k$, induced-subgraph superadditivity gives

$$s^+(G) \geq \sum_{i=1}^k s^+(G[V_i]). \quad (1)$$

This is the theorem of Akbari, Kumar, Mohar, and Pragada [1].

Write $T = C_3$, $P = C_5$, and

$$\delta = \sec(\pi/5) - 1 = \sqrt{5} - 2 < \frac{1}{2}, \quad \delta_q = \sec(\pi/q) - 1 < 1 \quad (q \equiv 1 \pmod{4}).$$

We use the following packet bounds. All allow arbitrary trees attached at arbitrary vertices, and mixed two-cycle packets allow arbitrary bridge connectors:

$$\sigma(T) > 0, \quad \sigma(P) \geq -\delta, \quad \sigma(C_q) \geq -\delta_q, \quad (2)$$

$$\sigma(TT) > 1, \quad \sigma(TP) > 1 - \delta, \quad \sigma(TC_q) > 1 - \delta_q. \quad (3)$$

Moreover,

$$\sigma(H) \geq 0 \quad \text{for every bicyclic or tricyclic cactus } H, \quad (4)$$

$$\sigma(H) > 0 \quad \text{for every tetracyclic or pentacyclic cactus } H. \quad (5)$$

For the indicated shared-cut clusters we use

$$\sigma(TTT) > 2, \quad \sigma(TTTT) > 3, \quad (6)$$

$$\sigma(TTP) > 2 - \delta \quad \text{when its two triangles share a cut,} \quad (7)$$

$$\sigma(TPP) > 6 - 2\sqrt{5}, \quad (8)$$

$$\sigma(PP) \geq 1 - \frac{4}{3\sqrt{13}}. \quad (9)$$

Bounds (2)–(8) and the lower-rank theorems are proved in the bicyclic, tricyclic, tetracyclic, and pentacyclic companion manuscripts [3, 4, 5, 6]; (9) is [3, Theorem 5.1]. If every cycle is 3 (mod 4) and the cycle-packing number is at most two, the Sachs half-plane theorem [8] gives, for an r -cyclic cactus,

$$\sigma(H) > r - 1. \quad (10)$$

This gives the shared TTT estimate. For four triangles it also applies unless the incidence is a central triangle with three disjoint triangular petals. In that sole case the exact grouped-Sachs matching injection in [8, Proposition 4.11] gives $s^+ > s^-$ and hence $\sigma(TTTT) > 3$. These are the only external packet dependencies.

2 Connector territories, intervals, and openings

Join two cyclic blocks in the *shared-cut graph* when they share a cut vertex, and call its connected components shared-cut clusters. In the block-cut tree contract the incidence subtree of each cluster to one marked node. Every cluster, including a singleton cyclic block, is marked. Take the minimal subtree spanning the marked nodes and suppress only unmarked degree-two nodes. The result is the *reduced cluster tree*.

Lemma 2.1 (Arbitrary connector territories). *Let $S_1, \dots, S_k$ be pairwise vertex-disjoint connected subtrees of the reduced cluster tree, and suppose every marked node belongs to exactly one S_i . Then $V(G)$ has a partition into connected induced territories $G_1, \dots, G_k$ such that G_i retains exactly the cycles assigned to S_i . Cuts occur on actual bridges, and every attached tree is assigned whole to one territory.*

Proof. Assign every remaining reduced-tree vertex to a nearest labelled subtree, breaking ties by a fixed order. In a tree each label class is connected. Every edge between classes expands to a path consisting only of bridge blocks; cut one actual bridge on it and assign the two remnants to their labels. At a Steiner branch, assign the branch vertex and one initial segment to one class and cut an actual bridge on every other incident segment. A component outside the cyclic hull has a unique attachment to it, for two attachments would make a cycle in the block-cut tree; assign that entire component to the owner of the attachment. The resulting sets are connected, induced, disjoint, and retain exactly the asserted cycles. $\square$

Lemma 2.2 (Intervals and private openings). *Let C be a cycle node of a cycle-cut incidence tree.*

1. *If the branches of $I - C$ are assigned to nonempty consecutive intervals of C , one interval per nonempty group of cyclically consecutive marks, these intervals and their branches form connected induced territories. Every mark has one owner and C is retained by none.*
2. *If v is a private vertex of C , put v and all off-cycle branches rooted there in F . Then F is a nonempty tree with $\sigma(F) = -1$. If $C - v$ contains all shared cyclic cuts needed by the other cycles, its complement is a connected induced territory.*

Both statements remain true with arbitrary hanging trees.

Proof. Cut the cycle edges between consecutive intervals. Every interval is a proper path, and every incidence branch follows its unique mark. For an opening, $C - v$ is a path and all branches rooted at v form a tree with it. Every nonempty tree F has $s^+(F) = |F| - 1$, so $\sigma(F) = -1$. Unique attachment of hanging trees proves the remaining assertions. Crossing edges are discarded only through (1); no edge monotonicity is used. $\square$

3 The exact DNN residual classification

For a connected edge-containing graph define

$$\kappa(G) = \sup_{\substack{M \succeq 0, M \geq 0 \\ \mathbf{1}^T M \mathbf{1} > 0}} \frac{4 \left(\sum_{uv \in E(G)} \sqrt{M_{uv}} \right)^2}{\mathbf{1}^T M \mathbf{1}}.$$

The sharp cactus DNN theorem [7, 9] gives, for bridge-block count b and cycle lengths $\ell_1, \dots, \ell_r$,

$$\kappa(G) = b + \sum_j \kappa(C_{\ell_j}), \quad \kappa(C_\ell) = \begin{cases} \ell, & \ell \text{ even}, \\ \ell \sec^2(\pi/(2\ell)), & \ell \text{ odd}, \end{cases} \quad s^-(G) \leq \kappa(G). \quad (11)$$

For completeness, fold each nonedge entry of a DNN matrix into its two diagonal entries by adding $M_{uv}(e_u - e_v)(e_u - e_v)^T$. Rotational averaging on a cycle and its least Fourier eigenvalue give the displayed cycle constants. Orthogonal gluing of folded Gram systems at a one-vertex sum, followed by Cauchy-Schwarz, proves block additivity. Finally, writing $A = A_+ - B$, the Schur product theorem makes $B \circ B$ DNN and gives

$$s^-/2 = - \sum_{uv \in E(G)} B_{uv} \leq \sum_{uv \in E(G)} |B_{uv}|, \quad \mathbf{1}^T (B \circ B) \mathbf{1} = s^-;$$

the definition of κ yields $(s^-)^2 \leq \kappa(G)s^-$.

Put

$$\epsilon_\ell = \begin{cases} 0, & \ell \text{ even}, \\ \ell \tan^2(\pi/(2\ell)), & \ell \text{ odd}. \end{cases}$$

Then

$$\epsilon_3 = 1, \quad \epsilon_5 = 5 - 2\sqrt{5} =: a, \quad 3a < 2, \quad 2a > 1, \quad \epsilon_5 + \epsilon_7 < 1. \quad (12)$$

These are exact: $3a < 2$ is $169 < 180$, and $2a > 1$ is $80 < 81$ after squaring positive sides. Also $\cos(\pi/7) > 1 - \frac{1}{2}(22/49)^2 > 7/8$, whence $\epsilon_7 < 7/15 < 2\sqrt{5} - 4 = 1 - \epsilon_5$; the middle inequality squares to $4489 < 4500$. The odd sequence decreases because, with $u = \pi/(2x)$, the derivative of $\tan^2 u/u$ is positive exactly when $2u > \sin u \cos u$.

For a hexacyclic cactus, block counting and (11) give

$$b + \sum_{i=1}^6 \ell_i = n + 5, \quad \sigma(G) \geq 5 - \sum_{i=1}^6 \epsilon_{\ell_i}. \quad (3.1)$$

If at most three cycles are triangles, the sum is at most $3 + 3a < 5$. With exactly four triangles, an even remaining length contributes zero; otherwise every pair except 5, 5 contributes at most $\epsilon_5 + \epsilon_7 < 1$. The pair 5, 5 survives because $2a > 1$. With at least five triangles the multiset is $\{3, 3, 3, 3, 3, q\}$, $q \geq 3$. Thus (3.1) is strict outside exactly

$$TTTTTQ = \{3, 3, 3, 3, 3, q\} \quad (q \geq 3), \quad TTTTPP = \{3, 3, 3, 3, 5, 5\}. \quad (13)$$

4 A five-triangle shared cluster

Lemma 4.1 (An intersecting-triangle $TTTP$ packet). *If T_1, T_2, T_3, P form one shared-cut cluster and two of the triangles share a cut, then $\sigma(TTTP) > 1$.*

Proof. Let k be the number of shared cyclic cuts on P . The incidence excess for four cycle nodes is three. The k cuts on P consume at least k units, and the intersecting triangle pair consumes another unit, either at a cut off P or as extra degree at a cut on P . Hence $k \leq 2$. Choose a private vertex v of P and open it as in Lemma 2.2. The path $P - v$ retains every cut on P , so the induced remainder H is connected and has exactly the three triangular cycles. The intersecting pair shows that their cycle-packing number is at most two. Hence the favorable-cycle theorem (10), which does not require one shared-cut cluster, gives $\sigma(H) > 2$. The opened tree has surplus -1 , and therefore $\sigma(TTTP) > 2 - 1 = 1$. $\square$

Lemma 4.2. *If five triangular blocks form one shared-cut cluster A , then $\sigma(A) > 2$, including the cycle-packing-three case.*

Proof. The bipartite incidence tree has a leaf triangle. It has one shared cyclic cut, so open one of its two private vertices by Lemma 2.2. The tree territory costs one and the remaining four triangles stay connected in one shared-cut cluster. Their surplus is greater than three by (6); this includes the central-triangle/three-petal incidence through the exact Sachs matching injection recalled above. Hence $\sigma(A) > 3 - 1 = 2$. In particular no packing-at-most-two assumption is hidden in the estimate. $\square$

5 Disconnected shared-cut graph: the $TTTTTQ$ family

Proposition 5.1. *If the cycles are $TTTTTQ$ and the shared-cut graph is disconnected, then $s^+(G) > |V(G)|$.*

Proof. If $q = 3$, the triangular block-graph theorem [8] applies. Assume $q \neq 3$. The minimal reduced cluster tree has at least two marked leaves, and at most one contains Q . Choose a Q -free leaf cluster A , containing r triangles, and cut the first actual bridge toward the rest.

For $r = 1, 2$, the remainder is respectively penta- or tetracyclic and is strictly positive by (5); for $r = 3, 4$ it is respectively tri- or bicyclic and is nonnegative by (4). The all-triangle side is strict, so all these sums are positive. If $r = 5$, Lemma 4.2 gives $\sigma(A) > 2$. The remote Q side is nonnegative unless $q \equiv 1 \pmod{4}$, and in that case it is at least $-\delta_q$ with $\delta_q < 1$. Thus the total is $> 2 - \delta_q > 1$. Lemma 2.1 makes this proof independent of connector shape and entry position. $\square$

6 Disconnected shared-cut graph: all 28 $TTTTTP$ partitions

Up to permutations of the four triangles and two pentagons, there are exactly 29 colored set partitions of $(4, 2)$: recursively choose the first nonzero pair (t, p) and require the remaining pairs to be lexicographically nondecreasing. Removing the one-cluster partition leaves the following 28 proper partitions. The table is both the complete list and the direct packet audit; the five named rows are proved immediately afterward.

partition	certificate
$T T T T P P$	Lemma 6.1
$TT T T P P$	$> 1 - 2\delta$ plus strict T 's
$TTT T P P$	$> 2 - 2\delta$ plus a strict T
$TT TT P P$	$> 2 - 2\delta$
$TTTT P P$	$> 3 - 2\delta$
$TP T T T P$	$> 1 - 2\delta$ plus strict T 's
$TTP T T P$	Lemma 6.3
$TT TP T P$	$> 2 - 2\delta$ plus a strict T
$TTTP T P$	Lemma 6.5
$TTT TP P$	$> 3 - 2\delta$
$TTP TT P$	$> 1 - \delta$
$TTTTP P$	Lemma 6.4
$PP T T T T$	shared $PP \geq 0$ plus strict T 's
$TT PP T T$	> 1 plus nonnegative PP and strict T 's
$TTT PP T$	> 2 plus nonnegative PP and a strict T
$TT TT PP$	> 2 plus nonnegative PP
$TTTT PP$	> 3 plus nonnegative PP
$TPP T T T$	$> 6 - 2\sqrt{5}$ plus strict T 's
$TP TP T T$	$> 2(1 - \delta)$ plus strict T 's
$TTPP T T$	positive tetracyclic packet plus strict T 's
$TTP TP T$	$> 1 - \delta$ plus a strict T
$TPP TT T$	$> 7 - 2\sqrt{5}$ plus a strict T
$TTTTPP T$	positive pentacyclic packet plus a strict T
$TT TP TP$	$> 3 - 2\delta$
$TTTP TP$	$> 1 - \delta$
$TTT TPP$	$> 8 - 2\sqrt{5}$
$TTPP TT$	positive tetracyclic packet plus > 1
$TTP TTP$	Lemma 6.2

Every direct quantity is positive because $\delta < 1/2$ and $6 - 2\sqrt{5} > 0$. Whole cluster territories are supplied by Lemma 2.1.

Lemma 6.1 (All singletons). *The partition $T|T|T|T|P|P$ has positive surplus for every reduced cluster tree.*

Proof. A triangular leaf gives a strict triangle and a positive pentacyclic remainder. Otherwise the two pentagons are all the leaves, so the finite tree has exactly two leaves and is a path. In path order form $TP + TT + TP$; its surplus is $> 2(1 - \delta) + 1 > 0$. $\square$

Lemma 6.2 (The double TTP row). *The partition $TTP|TTP$ has positive surplus.*

Proof. If the triangles in either cluster meet, (7) plus nonnegativity of the other tricyclic cluster gives a positive total. Otherwise each cluster is the distinct-cut chain $T - P - T$. Split each central pentagon into two proper intervals, assigning the bridge connector to the interval containing its actual entry. Join those selected intervals through the connector. The exact territories are $TT + T + T$, with surplus > 1 plus two strict terms. This coordinated split gives every connector vertex and shared cut one owner. $\square$

Lemma 6.3 (The $TTP|T|T|P$ row). *Every cactus with cluster partition $TTP|T|T|P$ has positive surplus.*

Proof. An intersecting pair in TTP gives $> 2 - \delta$, which with the remote pentagon gives $> 2 - 2\delta > 0$. A singleton triangular leaf gives a triangle and a positive pentacyclic remainder. Assume neither and write the cluster as the chain $T - P_0 - T$. If a reduced edge separates TP from a tetracyclic side, or TT from TPP , the corresponding packet sum is positive. If no such edge exists, the TTP cluster and remote pentagon are the only leaves; the two singleton triangles are internal on their path. Split P_0 according to its actual connector entry and form

$$TT + T + TP, \quad \sigma > 1 + 0 + (1 - \delta) > 0.$$

This also covers entries through a triangle, at its attachment cut, or at a private vertex of P_0 . $\square$

Lemma 6.4 (E1: the six-row $TTTTTP|P$ proof). *The partition $TTTTTP_0|P_1$ has positive surplus for every connector entry.*

Proof. If the four triangles are pairwise disjoint, incidence acyclicity makes P_0 a four-petal hub. An entry through a petal travels with that petal; an entry at a private hub vertex is paired with an adjacent petal mark. Splitting P_0 gives $TP + TTT$, hence $> 1 - \delta$.

Suppose two triangles meet. Delete P_0 from the five-cycle incidence tree. Every resulting component contains triangles, meets P_0 at one distinct cut, and is one component of the triangle shared-cut graph. At least one has size at least two, so the component-size partition and the size of the component containing the connector entry are exactly

sizes	entry size	certificate
(4)	4	two-pentagon opening
(3, 1)	3	$TTTTP + T$
(3, 1)	1	$TP + TTT$
(2, 2)	2	$TTP + TT$
(2, 1, 1)	2	$TTP + T + T$
(2, 1, 1)	1	$TP + TT + T$

This is exhaustive even for a multiway cut: all triangles at that cut lie in one component. For two or three components, cut P_0 in one gap between each consecutive pair of component marks. Give P_1 and its entire connector to the interval owning the marked entry component. The five displayed packet sums are positive by (2)–(5); in particular $TTTTP + T$ and $TTP + T + T$ are strict although their first term need only be qualitatively positive or nonnegative.

If instead the connector enters a private vertex z of P_0 , open P_0 at z and assign the connector and remote pentagon P_1 to the rooted territory at z . This territory is a pentagonal unicyclic cactus and has surplus at least $-\delta$. The complementary induced territory contains all four triangles; the path $P_0 - z$ retains every shared cut, and an intersecting pair ensures that its shared-cut cluster packet sum is greater than one. The total is therefore $> 1 - \delta > 0$.

For (4), open private vertices of both pentagons away from the connector route. Their tree territories cost two. The four triangles remain one shared-cut cluster with surplus > 3 , so the total is $> 3 - 2 = 1$. Entries on a triangle, at an off- P_0 triangle cut, at a multiway attachment cut, or through a rooted branch all belong to one of the six rows and have one owner. $\square$

Lemma 6.5 (E2: eight incidences and twenty-six entries). *The partition $TTTTP_0|T_4|P_1$ has positive surplus for every reduced-tree topology and every pair of labelled connector entries.*

Proof. If the T_4 – P_1 path avoids the $TTTTP_0$ cluster, use TP plus a positive tetracyclic territory. Otherwise its two arms enter that cluster. The exact colored incidence trees, by cut count $c = 1, 2, 3$, number

$$1, 3, 4.$$

With triangle nodes 0, 1, 2, pentagon node 3, and cut nodes starting at 4, their canonical edge sets are

$$\begin{aligned} c = 1 : & \quad ((0, 4), (1, 4), (2, 4), (3, 4)); \\ c = 2 : & \quad ((0, 4), (0, 5), (1, 4), (2, 4), (3, 5)), \\ & \quad ((0, 4), (0, 5), (1, 4), (2, 5), (3, 4)), \\ & \quad ((0, 4), (1, 4), (2, 5), (3, 4), (3, 5)); \\ c = 3 : & \quad ((0, 4), (0, 5), (0, 6), (1, 4), (2, 5), (3, 6)), \\ & \quad ((0, 4), (0, 5), (1, 4), (1, 6), (2, 5), (3, 6)), \\ & \quad ((0, 4), (0, 5), (1, 4), (2, 6), (3, 5), (3, 6)), \\ & \quad ((0, 4), (1, 5), (2, 6), (3, 4), (3, 5), (3, 6)). \end{aligned}$$

Seven contain a cut incident with two triangles; Lemma 4.1 gives $\sigma(TTTP_0) > 1$, which absorbs $\sigma(P_1) \geq -\delta$ independently of both entries. The eighth is the pentagon hub with three disjoint triangular petals.

Project an entry through a petal to its hub mark and force it to travel with that petal; project every other entry to its actual hub vertex. Thus three unlabelled distinct petal marks and ordered roots b (for T_4) and p (for P_1) lie on five hub vertices, with coincidences allowed. Modulo the dihedral group there are exactly 26 configurations. Exact enumeration of all nonempty proper cyclic intervals finds in every configuration an interval containing p which, after assigning T_4 according as b lies in it, owns one or two triangles. The certificate count is

$$20 : TP + TTT, \quad 6 : TTP + TT.$$

The totals are respectively $> (1 - \delta) + 0$ and $> 0 + 1$. Both interval and complement are nonempty consecutive sets; each petal mark and labelled arm has one owner. The executable certificate and its canonical incidences are specified in Appendix A. $\square$

Proposition 6.6. *If the cycles are $TTTTTP$ and the shared-cut graph is disconnected, then $s^+(G) > |V(G)|$.*

Proof. The twenty-eight-row table is the complete colored cluster partition list. Twenty-three rows follow directly from the packet ledger; the remaining five are Lemmas 6.1–6.5. Thus every reduced-tree topology, connector entry, and colored partition is covered. $\square$

7 Fully shared $TTTTTQ$: exact census

For one shared-cut cluster let I be its bipartite cycle–cut incidence tree. With six cycle nodes and c cut nodes,

$$|E(I)| = c + 5, \quad \sum_{x \text{ cut node}} (\deg_I(x) - 1) = 5, \quad 1 \leq c \leq 5. \quad (14)$$

Triangle degrees are at most three; a q -cycle degree is at most q .

Lemma 7.1 (The $TTTTTQ$ census). *Fix Q and quotient by permutations of the five triangles and cut nodes. The numbers of admissible incidence trees are*

Q capacity	$c = 1$	$c = 2$	$c = 3$	$c = 4$	$c = 5$	total
$q = 3$	1	6	20	27	14	68
$q = 4$	1	6	20	28	15	70
$q \geq 5$	1	6	20	28	16	71

Every nonbouquet tree admits a positive ordinary one-cycle interval split: respectively 67, 69, 70 trees. The sole exception in every regime is

$$((0, 6), (1, 6), (2, 6), (3, 6), (4, 6), (5, 6)), \quad (15)$$

where $0, \dots, 4$ are triangles, $5 = Q$, and 6 is the common cut.

Proof. Enumerate cut neighborhoods with $c + 5$ incidences, reject zero cycle degree, cut degree below two, excess capacity, and every non-tree, and canonicalize by color-preserving permutations. This gives the displayed counts. For every cycle of incidence degree at least two, delete its node and inspect the branch multisets. Splitting Q leaves strict all-triangle branches. Splitting a triangle leaves one Q -branch: TQ is positive; generic TTQ is nonnegative and another triangle branch is strict; larger T^kQ branches are lower-rank positive. If Q is singleton, the other four triangles occupy at most two branches, so one contains TT and its surplus > 1 absorbs the hostile loss $\delta_q < 1$. These tests certify every nonbouquet tree. The executable enumeration and asserted canonical exception reproduce the finite step exactly. $\square$

Proposition 7.2. *If all six $TTTTTQ$ cycles form one shared-cut cluster, then $s^+(G) > |V(G)|$.*

Proof. Realize each split certified by Lemma 7.1 with consecutive intervals as in Lemma 2.2; the branch packet sum is positive. For the bouquet, open a private vertex of Q and a private vertex of one designated triangle. The path remnants retain the common cut, and the other four triangles remain one shared-cut cluster. Thus

$$\sigma(G) > 3 - 1 - 1 = 1.$$

This also covers $Q = T$: the designation is arbitrary. The regimes $q = 3$, $q = 4$, and $q \geq 5$ are exhaustive by (14). $\square$

8 Fully shared $TTTTTP$: exact 150-tree census

Lemma 8.1 (The SAFE incidence census). *Modulo permutations of four triangles, two pentagons, and cut nodes, the exact counts are*

c	1	2	3	4	5	total
incidence trees	1	9	40	62	38	150
SAFE ordinary split	0	9	40	62	37	148

The two canonical exceptions, with $0, 1, 2, 3 = T$, $4, 5 = P$, and cuts beginning at 6, are

$$((0, 6), (1, 6), (2, 6), (3, 6), (4, 6), (5, 6)); \quad (16)$$

$$((0, 6), (4, 6), (1, 7), (4, 7), (2, 8), (4, 8), (3, 9), (4, 9), (4, 10), (5, 10)). \quad (17)$$

They are respectively the six-cycle bouquet and a saturated pentagon hub with four triangular petals and one pentagonal petal.

Proof. Equation (14), cut degree at least two, and cycle capacities define the finite enumeration. Canonicalization uses $S_4 \times S_2 \times S_c$. For each of all $6 \cdot 150 = 900$ cycle choices, delete the cycle node and compute the retained incidence components. The SAFE ledger is exactly

branch	checked retained hypothesis	bound
T	none	> 0
P	none	$\geq -\delta$
TT	none	> 1
TP	none	$> 1 - \delta$
PP	one shared-cut component	$\geq 1 - 4/(3\sqrt{13})$
TTT	one shared-cut cluster	> 2
TPP	one shared-cut cluster	$> 6 - 2\sqrt{5}$
TTP	the triangles retain a common cut	$> 2 - \delta$
other rank three	none	≥ 0
$TTTT$	one shared-cut cluster	> 3
$TTTP$	an intersecting triangle pair	> 1 by Lemma 4.1
other rank four	none	> 0
rank five	none	> 0

A split is accepted only if the corresponding exact symbolic bound is positive, or equals zero with a strict summand. In particular, qualitative lower-rank positivity is never used to cancel a negative singleton pentagon, and concentrated estimates are applied only after checking their retained cuts. The asserted totals are 1, 9, 40, 62, 38, the accepted totals are 0, 9, 40, 62, 37, and exhaustive canonicalization leaves exactly (16) and (17). Appendix A gives the executable certificate. $\square$

Proposition 8.2. *If all six $TTTTPP$ cycles form one shared-cut cluster, then $s^+(G) > |V(G)|$.*

Proof. For each of the 148 SAFE trees, realize the certified split by proper consecutive intervals and apply its positive packet sum. In the bouquet, open private vertices on both pentagons. Their path remnants retain the common cut, the four triangles remain one shared-cut bouquet, and

$$\sigma(G) > 3 - 1 - 1 = 1.$$

In the saturated hub, the hub pentagon uses all five vertices. In cyclic order, merge the pentagonal-petal mark with either neighboring mark; both neighbors are triangular. Give the remaining three triangular marks separate proper intervals. This produces

$$TP + T + T + T, \quad \sigma(G) > (1 - \delta) + 0 + 0 + 0 > 0,$$

with every strict triangle retained. Thus both exact exceptions are covered. $\square$

9 Main theorem

Theorem 9.1. *Every connected hexacyclic cactus G on n vertices satisfies*

$$\boxed{s^+(G) > n}.$$

Proof. The exact DNN estimate (3.1) is strict unless the cycle multiset is $TTTTTQ$ or $TTTTTP$. If the shared-cut graph is disconnected, Propositions 5.1 and 6.6 apply. If it is connected, all cycles

form one incidence-tree cluster and Propositions 7.2 and 8.2 apply. These alternatives exhaust every cycle multiset and every incidence. Lemmas 2.1 and 2.2 include arbitrary connectors, Steiner branches, entry vertices, and attached trees. Every comparison is on a vertex partition into induced subgraphs; no edge monotonicity or tree relocation is used. $\square$

Since a connected hexacyclic graph has $|E(G)| = n + 5$, this proves the strict conclusion of the Akbari–Kumar–Mohar–Pragada–Zhang conjecture [2, Conjecture 1.2] for every hexacyclic cactus.

A Exact reproduction and canonical proof objects

From the repository root run, using only the Python standard library,

```
python research/hexacyclic-e2-tttp-entry-census.py
python research/hexacyclic-tttttq-incidence-census.py
python research/hexacyclic-ttttpp-incidence-census.py
```

The first script asserts the *TTTP* incidence counts {1: 1, 2: 3, 3: 4}, seven intersecting-pair trees, the unique three-petal hub, and exactly

$$26 = 20 (TP + TTT) + 6 (TTP + TT)$$

ordered labelled-entry orbits. It prints a canonical incidence, interval, and packet certificate for each orbit.

The second script asserts

```
q=3: {1: 1, 2: 6, 3: 20, 4: 27, 5: 14} total 68
q=4: {1: 1, 2: 6, 3: 20, 4: 28, 5: 15} total 70
q>=5: {1: 1, 2: 6, 3: 20, 4: 28, 5: 16} total 71
```

and the sole canonical exception (15) in each regime. The third asserts

```
colored trees by cut count: {1: 1, 2: 9, 3: 40, 4: 62, 5: 38}
SAFE one-cycle packet resolutions: {2: 9, 3: 40, 4: 62, 5: 37}
```

together with all 900 cycle choices, the exact retained-incidence ledger, and the two canonical exceptions (16) and (17). The scripts enumerate finite incidence objects; the spectral implications are the symbolic packet lemmas proved or cited in the manuscript. The last executable uses only exact **Fraction** arithmetic. It replaces radical margins by the rational consequences $\delta < 1/4$, $1 - \delta > 3/4$, $2 - \delta > 7/4$, $6 - 2\sqrt{5} > 3/2$, and $1 - 4/(3\sqrt{13}) > 0$, each certified by squaring positive rational bounds. Thus no numerical approximation is a mathematical premise.

The source proof objects integrated into this manuscript are

```
research/hexacyclic-dnn-residuals-2026-07-26.md
research/five-triangle-shared-cluster-surplus-2026-07-26.md
research/hexacyclic-tttttq-disconnected-2026-07-26.md
research/hexacyclic-ttttpp-disconnected-audit-2026-07-26.md
research/hexacyclic-ttttpp-e1-resolution-2026-07-26.md
research/hexacyclic-e2-tttp-two-entry-resolution-2026-07-26.md
research/hexacyclic-tttttq-incidence-census.py
research/hexacyclic-ttttpp-incidence-census.py
research/hexacyclic-e2-tttp-entry-census.py
```

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